

Tide retrospective: second-order moment rates (gamma-rung programme, stage 3)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Stage 2 gave the second-order J_n expansions; this stage converts them into rates for the Gibbs moments themselves, the direct inputs of the fourth-cumulant assembly. The scoping consult's worked expansions fixed every target coefficient in advance, and all three were confirmed numerically before any Lean was written.

2 The thing formalised

Ten declarations in `Laplace/OneD/MomentSecondOrder.lean`, sorry-free with zero warnings: three bridge identities ($t\langle x^2 \rangle = J_2/(\lambda J_0)$ and the x^3, x^4 analogues, from the seabed's substitution relation), four J -deltas with the Gaussian constants evaluated, and the three public rates. With A, B the cubic and quartic scales, eventually in t :

$$\left| t\langle x^2 \rangle - \frac{1}{\lambda} - \frac{45A^2 - 12B}{\lambda t} \right| \leq \frac{K}{t\sqrt{t}}, \quad \left| t^2\langle x^4 \rangle - \frac{3}{\lambda^2} - \frac{450A^2 - 96B}{\lambda^2 t} \right| \leq \frac{K}{t\sqrt{t}},$$
$$\left| t^2\langle x^3 \rangle + \frac{15A}{\sqrt{\lambda}^3} \right| \leq \frac{K}{\sqrt{t}}.$$

Together with the seabed's existing mean rate, these are exactly the inputs the κ_4 cancellation table consumes.

3 Proof strategy

Each rate is a fraction analysis over the J -form bridge: combine the difference into a single fraction with numerator $N = J_r - (\text{leading} + \text{coeff}/t) J_0$; split N against the two deltas, where the $1/t$ terms cancel *exactly* ($q_2 - q_0 = c_2$ and $p_4 - 3q_0 = c'_4$, verified by `ring` on explicit coefficients); bound the three pieces by the triangle inequality; and control the denominator by the J_0 lower bound. The third moment is the same shape one order earlier, with odd Gaussian moments annihilating everything but the single A -term.

4 Roadblocks and resolutions

Built across four checkpointed commits (bridges, deltas, first assembly, remaining two); the deltas layer compiled on the first attempt. Every iteration was a documented catalogue class — including,

satisfyingly, one instance of the unsafe t -inside-radical rewrite that this programme's own stage-2b `CLAUDE.md` entry warns about, caught immediately from the goal display ($\sqrt{\sqrt{t^2}}$) and repaired with the radical-atom pattern. New micro-entries: `div_le_div_iff` and `le_div_iff` are now the $_0$ -suffixed variants, and `set_option` must precede a declaration's docstring.

5 What was Mathlib, what was new

Mathlib: field and absolute-value algebra, `gcongr`. Seabed: the substitution relation, the J_0 bounds, and the whole stage-2 layer. New: the exact-cancellation fraction pattern, now used three times and likely the template for any future order-deepening.

6 Lessons

Checkpointed WIP commits (each building, sorry-free) made a four-hundred-line file safe to build across session boundaries; the per-layer validation (bridges, then deltas, then assemblies) meant no error ever implicated more than the newest layer. When a catalogue entry exists, the failure it describes costs one iteration; the only multi-iteration fights in this stage were in the first assembly, whose structure the other two then reused verbatim.

7 Follow-ups

- Stage 4–5: the κ_4 assembly over these rates and the existing mean rate, and the limit $t^3\kappa_4 \rightarrow -\gamma/\lambda^4 + 3\alpha^2/\lambda^5$.
- Stage 6: the gamma recovery corollary (uniqueness of limits), completing the anharmonic jet $(\lambda, \alpha, \gamma)$ and earning the germbij note's Theorem 3.1 its second marker.